

DRISHYAM

From Footage to Action

Software Requirements Specification (SRS)

Functional & Non-Functional Requirements

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1. Introduction

This document specifies the functional and non-functional requirements for the Drishyam real-time AI surveillance platform. It serves as the contract between product and engineering.

2. Functional Requirements

2.1 Camera & Stream Management

- FR-CAM-01: System shall support RTSP and ONVIF camera discovery/addition.
- FR-CAM-02: System shall maintain continuous health status (online/offline/degraded) for each camera.
- FR-CAM-03: System shall allow configuration of analysis resolution and FPS per camera or site.
- FR-CAM-04: System shall support at least 50 concurrent streams per edge node (MVP target).
- FR-CAM-05: System shall allow defining polygonal detection zones on camera views.

2.2 AI Detection Engine

- FR-AI-01: System shall detect the following classes in real time: Weapon (firearm/knife), Violence/Fight, Intrusion (zone breach), Loitering.
- FR-AI-02: Each detection shall carry a confidence score (0–1).
- FR-AI-03: System shall apply temporal filtering (minimum frames / duration) to reduce transient false positives.
- FR-AI-04: System shall generate a short evidence clip (configurable 5–15 seconds) around the detection timestamp.
- FR-AI-05: System shall support per-site / per-camera enable/disable of detection classes and confidence thresholds.
- FR-AI-06: System shall preferentially run inference on edge devices; fall back to cloud only when necessary.

2.3 Alerting & Notification

- FR-ALT-01: On confirmed detection, system shall create an Alert record with severity, type, camera, timestamp, confidence, and clip reference.
- FR-ALT-02: System shall push real-time notification via WebSocket to all eligible online users.
- FR-ALT-03: System shall send push notification (FCM/APNs) to mobile devices of eligible users.
- FR-ALT-04: Alert severity levels: Critical (weapon/violence), High (intrusion), Medium (loitering), Low (info).
- FR-ALT-05: Users shall be able to Acknowledge, Escalate, Mark False Positive, or Resolve an alert.
- FR-ALT-06: System shall record time-to-acknowledge and time-to-resolve metrics.

2.4 User & Organization Management

- FR-USR-01: Multi-tenant hierarchy: Organization → Site(s) → Camera(s).
- FR-USR-02: Roles: SuperAdmin (platform), OrgAdmin, SiteManager, Operator, Viewer.
- FR-USR-03: Permissions are role-based and optionally scope-limited to specific sites.
- FR-USR-04: OrgAdmin can invite users by email and assign roles.
- FR-USR-05: All authentication actions and privilege changes are audit-logged.

2.5 Dashboard & Visualization

- FR-UI-01: Live camera grid view with status indicators.
- FR-UI-02: Real-time alert feed with filtering by severity, site, type, status.
- FR-UI-03: Single alert detail view with video clip player, timeline, and action buttons.
- FR-UI-04: Basic reporting: alerts over time, false-positive rate, response times.

3. Non-Functional Requirements

3.1 Performance

- NFR-PERF-01: End-to-end detection latency (frame → alert) ≤ 5 seconds for 95th percentile under normal load.
- NFR-PERF-02: Alert notification delivery ≤ 1 second after alert creation for online WebSocket clients.
- NFR-PERF-03: Dashboard initial load ≤ 3 seconds on standard broadband.
- NFR-PERF-04: System shall support at least 200 concurrent cameras across edge nodes in MVP architecture.

3.2 Reliability & Availability

- NFR-REL-01: Target uptime 99.5% for core ingestion + detection path.
- NFR-REL-02: Camera disconnection shall be detected within 30 seconds and surfaced as health event.
- NFR-REL-03: Alert and clip data shall be durably stored; no silent loss of Critical alerts.

3.3 Security

- NFR-SEC-01: All API traffic over TLS 1.2+.
- NFR-SEC-02: Authentication via JWT (access + refresh) or session tokens.
- NFR-SEC-03: Role-based access control enforced on every protected endpoint.
- NFR-SEC-04: Secrets (DB, API keys, model endpoints) stored in environment / secret manager, never in code.
- NFR-SEC-05: Input validation and output encoding to prevent injection attacks.
- NFR-SEC-06: Clip access requires authentication and authorization; signed short-lived URLs preferred.

3.4 Privacy & Compliance

- NFR-PRI-01: Face blur option available for stored clips (configurable per organization).
- NFR-PRI-02: Data retention configurable; default 30 days for clips.
- NFR-PRI-03: Design aligned with DPDP principles (purpose limitation, minimal retention, auditability).
- NFR-PRI-04: Prefer edge processing to keep raw video local when possible.

3.5 Scalability

Architecture shall allow horizontal scaling of edge inference nodes and backend services independently. Database and object storage shall support growth to tens of thousands of cameras over time.

4. Roles & Permissions Matrix (Summary)

Action	OrgAdmin	SiteMgr	Operator	Viewer
Manage Org / Users	Yes	No	No	No
Add/Edit Cameras	Yes	Own sites	No	No
Configure Detection Rules	Yes	Own sites	No	No
View Live + Alerts	Yes	Yes	Yes	Yes
Acknowledge / Resolve Alerts	Yes	Yes	Yes	No
Mark False Positive	Yes	Yes	Yes	No
View Audit Logs	Yes	Limited	No	No

5. Error Handling Principles

- All API errors return structured JSON with code, message, and optional details.
- Camera stream failures generate health events, not silent drops.
- AI inference failures on a single frame/stream do not crash the pipeline; they are logged and skipped.
- Critical alerts that fail to deliver push notification are retried with exponential backoff.
- User-facing errors are human-readable; internal details go to logs only.

6. Validations

- Camera RTSP URL format and reachability checked on add/update.
- Email format and uniqueness within organization for invites.
- Role assignment only to valid roles for the tenant.
- Detection zone coordinates must form a valid polygon within frame bounds.
- Confidence threshold inputs clamped to [0.1, 0.99].